CivicEcho - A03

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01

Secure Document Format

Secure Document Format

- Hybrid encryption (AES-GCM + RSA)
- Confidentiality and integrity by design
- Authenticated metadata (AAD)
- Anonymity preserved

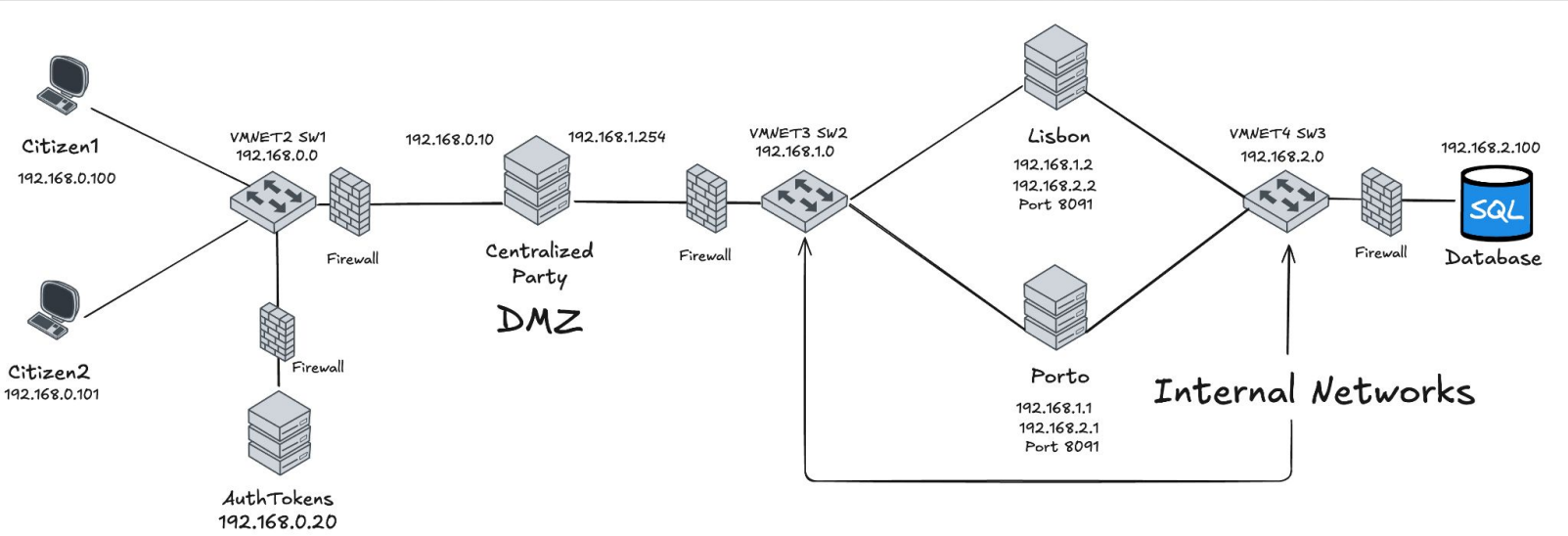
```
{
  "ciphered": "base64_encrypted_content",
  "iv": "base64_initialization_vector",
  "aad": {
    "timestamp": "report_timestamp",
    "location": "report_location",
    “operation”: “CREATE/READ”,
    “token”: “token + signature”
  },
  "key": "base64_rsa_encrypted_aes_key"
}
```

02

Built Infrastructure (Cybersecurity View)

Built Infrastructure

- Segmented network (3 subnets)
- DMZ + internal isolation
- Default-deny firewalls and protection against DoS attacks
- Minimal database exposure



03

Secure Channels and Key Distribution



Secure Channels and Key Distribution

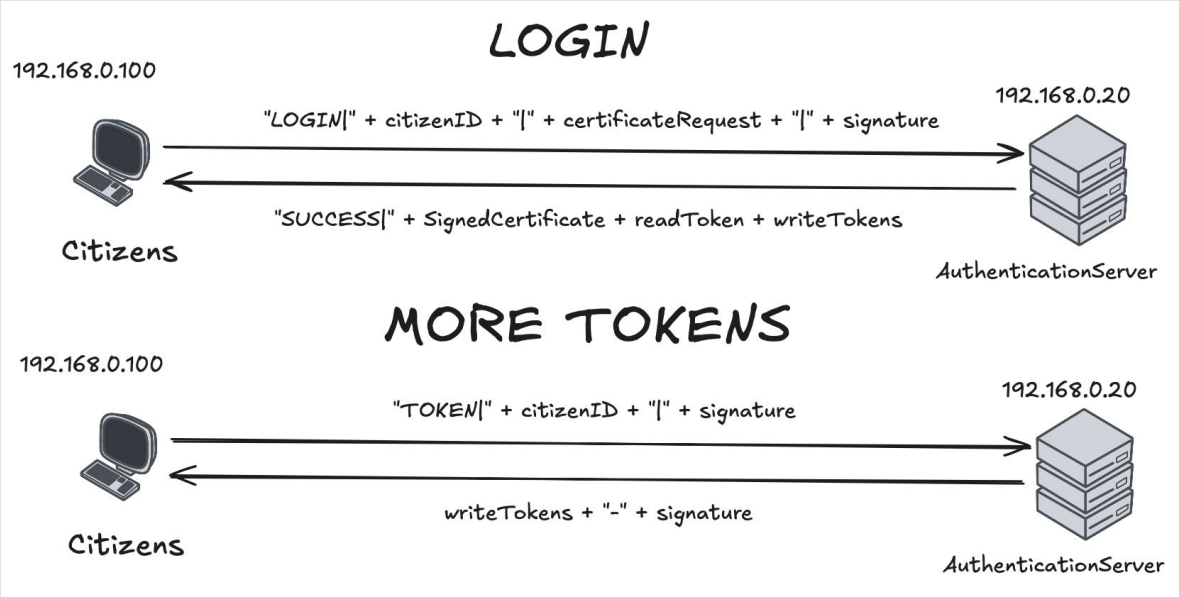
- TLS 1.3 everywhere
- cipher suite - TLS_AES_128_GCM_SHA256
- Certificate Authority rooted at Authentication Server
- Mutual authentication, except in login process
- Provides: confidentiality, authentication and non-repudiation



04

Security Challenge

Security Challenge



Security Challenge

- Write Token: random 256-bits, expiration time and citizen’s municipality. (Single-use)
- Read Token: random 256-bits and citizen’s municipality. (Eternal)
- Signed by Authentication Server
- Verified by Centralized Party
- Replay and spam prevention
- Location-bound authorization



05

Main Results and Security Guarantees

Main Results and Security Guarantees

- Confidentiality end-to-end
- Strong anonymity
- Integrity and non-repudiation
- Attacker model validated
- Future work - misinformation detection